Chirag Shah

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Profile

Electronics engineer with a track record of building and delivering technology from the ground up.
Developed India's first patented Drone-in-a-Box solution, led the development of a 720 kWh BESS project, and owned full-stack electronics development across multiple deep-tech startups. Currently developing drone electronics at Aereo.
Co-inventor on 6 patents filed at Log9 Materials across fuel cell management, power management, and system design. Named inventor on the SkyDock patent, granted across the US, European Patent Office, Australia, Denmark, and Finland (US-11772819-B2).

Visit my website chiragrshah.com where I have collated all my work.

Work Experience

Hardware Design Engineer - Aereo (Oct/25 – Present)

- Drove development of DC-DC converters, battery management systems (BMS), power distribution boards and autopilot boards
- Led end-to-end design assurance, reviewing and validating schematics, PCB layouts, and system architectures across all major hardware blocks prior to assembly
- Prevented multiple hardware failures by identifying critical design flaws in schematic design during pre-production reviews

Log9 Materials Scientific/Jupiter Electric Mobility (Aug/22 – Oct/25)

Deputy Engineer - Jupiter Electric Mobility (Oct/24 – Oct/25)

Log9 was acquired by Jupiter Electric Mobility (JEM). JEM designs and manufactures battery energy storage system (BESS), commercial 4W EVs and battery packs for railway applications.

- Managing the technical specifications and complete electrical systems integration (BMS, EMS, HVAC, Switchgear) for a 720kWh BESS project
- Collaborating with vendors (BMS, EMS, PCS, switchgear) and led the BESS integration/testing
- Researched Energy Management Systems (EMS) & BESS topologies for different use cases and tender specifications
- Led integration for a 20kWh BESS system for Indian Railways

Associate Electronics Engineer - Log9 Materials (Aug/22 – Oct/24)

Log9 was a nanotechnology company pioneering li-ion chemistry and battery pack manufacturing.
I was a core R&D team member in the **Aluminum Fuel Cell (AFC)** and **Battery Management System (BMS)** teams.

BMS team: Firmware Development (Jan/24-Oct/24)

- Supported BMS deployment for two 3W-OEMs
- Firmware development for new features and troubleshooting
- Integration and testing the BMS and battery pack for two 3W-OEMs
- Thermal management system prototype

AFC team: Electronics System Development (Aug/22 – Dec/23)

- Led electronics system development with a team of 3 engineers
- Specification definition and PCB design reviews for fuel cell management system (FMS), power management system (PMS) and slave FMS
- Firmware development on STM32 for the FMS, PMS, slave FMS
- Electronics and power electronics integration and testing
- Filed **six patents** in the areas of fuel-cell management, power management and fuel-cell system design

Embedded Design Engineer - Sagar Defense Engineering (May/19 – Jul/22) – Pune

SkyDock: Electronics and System Development (May/19 – Jun/21)
SkyDock is a launching and charging station which keeps the UAV in ready-to-fly condition 24/7. After landing it automatically starts charging the UAV. Our work was **granted a patent** by US, Denmark, European Patent Office, Australia and Finland (US-11772819-B2)

My responsibilities as a core R&D team member included

- Conceptualized the prototype, designed, and built hardware/firmware for all electrical/electronic systems
- Designed the complete PCB set (switching DC-DC converter, battery management system, charging management system etc)
- End-to-end assembly of the PCBs, BOM generation, vendor negotiations for procurement
- Led a team of 3-4 people to produce SkyDock

Other Projects: Assembly & PCB Design

- Set up the SMT assembly process – pick and place machine programming, solder paste handling and printing, reflow profiling, cleaning and inspection
- Assembled PCBs for UAV subsystems such as flight controllers, power distribution, gimbal control etc
- Designed and assembled various PCBs for power conversion, communication interfaces, and control modules

Internships – (2015 – 2019)

ideaForge: Design & Build of Engineering Validation Prototype (10/Dec/2018 – 18/Jan/2019)

- Designed the CAD model on Fusion 360, fabricated the parts (3D printing/CNC) and assembled the prototype
- Designed and built the electronics setup and firmware to drive the mechanical assembly

Fractal Analytics: Implementation of Room Occupancy System (11/Jun/2018 - 13/Jul/2018)

- Occupancy monitoring system across 9 meeting rooms running 24x7
- Designed to consume a low standby current (88µA) for battery operation. The devices connect to each other via a network of RF trans-receivers. Data was sent to AWS IOT core and then pulled into DynamoDB

e-Yantra IIT Bombay: Formation Control of Multiple Swarm Robots (22/May/2017 - 7/Jul/2017)

- 7 weeks residential internship at the Embedded and Real-Time Systems Lab advised by **Dr Kavi Arya, IIT Bombay** under the e-Yantra Summer Internship 2017 program
- Developed algorithms and embedded-C firmware to control swarm robots

Projects/Achievements

Coursera: Algorithms for Battery Management Systems Specialization: Oct 2020

5-Course Specialization dealing with batteries, battery management systems (BMS) and algorithms for BMS

Constant Current Load PCB: 2018

A microcontroller-based load which draws a programmable constant value of current to test power sources.

e-Yantra Robotics Competition IIT Bombay 2016: 1st Place (Out of 167 teams)

Designed and built the robotic arms; programmed the Firebird-V robot in embedded-C (ATmega-2560)

Technical Skills

- Embedded systems design & development
- Schematic design and circuit design
- PCB and flex PCB designing and assembly
- Embedded-C programming (AVR, STM32)
- Communication protocols: CAN, UART, SPI, USB
- Switching and linear power supply design
- BOM generation and procurement
- Analog & digital circuit design (ADC, Op-amps, MOSFETs)
- CAD (Fusion 360) and 3D printing

Education

2015 – 2019	Electronics Engineering - Sardar Patel Institute of Technology	8.37 CGPA
2013 / 2015	SSC / HSC	87.5% / 82.31%

Other Interests

- Certified PADI advanced open water scuba diver
- Sailing and windsurfing